OM AMAR KARANDE

AI / GenAI Engineer | Pune, Maharashtra, India | 9145097303 | omkarande2277@gmail.com
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PROFESSIONAL SUMMARY

AI/GenAI Engineer with hands-on experience shipping LLM, RAG, agentic and computer-vision systems using Python, LangChain, LangGraph and FastAPI. Built production-style document extraction, OCR and AI video generation pipelines for real client use cases, including privacy-first on-premise LLM stacks (Ollama, ChromaDB). Strong ML/DL foundation (PyTorch, TensorFlow, Scikit-learn). Seeking an AI Engineer / GenAI Developer role at a product-based company to build reliable, scalable AI products.

TECHNICAL SKILLS

Languages: Python, C
GenAI & LLMs: RAG, LangChain (LCEL), LangGraph, Agentic AI, Prompt Engineering, Tool Calling / Structured Output, Multimodal (Vision) LLMs, Ollama (Llama 3), Claude API, Gemini API, ChromaDB, Embeddings, ComfyUI
ML / DL / CV: PyTorch, TensorFlow, Scikit-learn, Computer Vision, NLP, OCR (PaddleOCR, Tesseract), Feature Engineering, Model Evaluation
Backend & Tools: FastAPI, Streamlit, Docker, Git/GitHub, Pandas, NumPy, PyMuPDF, n8n, Power Platform, Power BI, Jupyter, Antigravity IDE

PROFESSIONAL EXPERIENCE

AI Developer | AI India Innovations Pvt. Ltd. **Current**

- Developed Generative AI solutions with Python, LLMs, LangChain and AI APIs for client business use cases.
- Built RAG-based workflows and applied prompt engineering to improve output relevance and consistency.
- Produced AI video generation workflows, engineering prompts to keep visual consistency across multi-segment clips; benchmarked multiple video models and delivered a technical research report.
- Built API-based services applications.

PROJECTS

PDF RAG Assistant | *Python, LangChain LCEL, ChromaDB, Ollama, Streamlit*

- Built a 100% on-premise, privacy-first document Q&A system with LangChain LCEL, ChromaDB and Streamlit, so no document data ever leaves the machine.
- Indexed PDFs with Nomic embeddings and used a local Llama 3 8B model (Ollama) to generate answers grounded in the source documents.

P&ID Engineering Data Extractor | *Python, Gemini & Claude Vision APIs, PyMuPDF, Streamlit*

- Built a Streamlit app that uses multimodal vision LLMs to extract structured components from P&ID/PFD engineering drawings, with one-click Excel export.
- Ensured reliable structured output through schema-driven design (tool calling and response schemas) and high-resolution PDF rasterization with PyMuPDF.
- Implemented an in-context learning loop that reuses user corrections to continuously improve extraction quality.

Invoice OCR Processing | *Python, PaddleOCR, Tesseract, Regex, LLMs*

- Built an enterprise-grade Indian GST invoice OCR and document AI pipeline using dual-engine consensus (PaddleOCR + Tesseract) for robust text extraction.
- Designed self-healing regex parsers to extract structured invoice fields, validated with statistical RRSI benchmarks to measure accuracy.
- Developing an evaluation-driven self-improvement framework with experiment tracking, data leakage screening and cost-aware acceptance criteria.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning | CGPA: 8.4 2023 - 2027
D. Y. Patil Agriculture and Technical University, Talsande

Diploma in Electronics and Telecommunication Engineering 2020 - 2023
DKTE's Yashwantrao Chavan Polytechnic, Ichalkaranji

CERTIFICATIONS

Microsoft Certified: Power Platform Fundamentals (PL-900)
IBM: Python for Data Science, AI and Development